

CAPA Recommendation System Documentation

1. Introduction

In pharmaceutical, biotechnology, and medical-device industries, maintaining regulatory compliance and product quality is critical. Organizations operating under FDA, GMP, GxP, and ISO regulations must effectively manage deviations, non-conformances, audit findings, and quality incidents.

CAPA (Corrective and Preventive Action) is one of the most important quality-management processes used in regulated industries. The proposed AI-Powered CAPA Recommendation System assists quality teams using Machine Learning, NLP, semantic retrieval, and recommendation systems to intelligently suggest root causes, corrective actions, preventive actions, severity levels, and similar historical CAPA records.

2. Problem Statement

Traditional CAPA systems face several operational challenges:

- Manual investigation processes
- Time-consuming analysis of historical records
- Inconsistent recommendations
- Poor knowledge reusability
- Limited intelligent decision support
- Regulatory pressure requiring transparency and traceability

3. Proposed Solution

The proposed solution is a hybrid AI-based CAPA Recommendation System that combines:

- Multi-output classification
- Semantic similarity search
- Historical CAPA retrieval
- Recommendation ranking
- Interpretable recommendation techniques

The system acts as a decision-support platform for compliance and quality teams.

4. System Architecture

Workflow Architecture:

User Input → NLP Embedding → Multi-Output Classification → Similarity Search → Historical CAPA Retrieval → Recommendation Engine → Final Recommendations

5. Workflow

Step 1: User submits incident description.

Step 2: System converts text into embeddings using Sentence Transformers.

Step 3: Multi-output classifier predicts root cause, corrective action, preventive action, and severity.

Step 4: FAISS/ChromaDB retrieves similar historical CAPA records.

Step 5: Recommendation engine ranks relevant actions.

Step 6: Final recommendations and interpretability insights are displayed.

6. Technology Stack

Layer	Technology
Frontend	Streamlit
Backend API	FastAPI
Programming Language	Python
NLP Embeddings	Sentence Transformers
ML Classification	Scikit-learn / XGBoost
Vector Database	FAISS / ChromaDB
Structured Database	PostgreSQL
Data Processing	Pandas, NumPy
Interpretable Recommendations (Optional)	SHAP
Deployment	Docker, Render/AWS

7. Example Scenario

Input Incident:

“Repeated microbial contamination detected in sterile filling area.”

Predicted Outputs:

- Root Cause: Environmental Control Failure
- Corrective Action: Deep Sanitation
- Preventive Action: Airflow Monitoring
- Severity: Critical

Retrieved Similar CAPAs:

- CAPA-1045
- CAPA-1102

Recommended Actions:

- Perform deep sanitation
- Requalify HVAC systems
- Increase environmental monitoring frequency
- Conduct operator retraining

8. Advantages

- Faster investigations
- Improved consistency
- Knowledge reuse
- Transparent and interpretable recommendations
- FDA/GxP aligned architecture
- Scalable enterprise-ready design

9. Future Enhancements

- Knowledge graph integration
- Continuous learning using human feedback
- Advanced risk prediction
- Enterprise QMS integration

10. Conclusion

The AI-Powered CAPA Recommendation System transforms traditional CAPA management into a modern intelligent decision-support process. By combining Machine Learning, NLP, semantic retrieval, and interpretable recommendation techniques, the solution enhances operational efficiency, consistency, compliance readiness, and organizational knowledge reuse in regulated industries.